■ Mushroom Business Management & E-Commerce Platform (SporeERP)

Document 10: Master Product Requirements Document & Complete Technical Specification

Project Title: SporeERP (Mushroom Business Management & E-Commerce Platform)
Document Version: 1.0.0-FINAL-ENTERPRISE
Release Date: October 2026
Technology Stack: Next.js 15+ (App Router) * TypeScript * MariaDB 11.x * Prisma ORM * NextAuth.js v5 * Redis * TailwindCSS
Author & Governance: 2-Founder Executive Board & Lead Engineering Architect

1. Executive Summary & Product Architecture

SporeERP is an enterprise-grade, full-stack application designed specifically to power the end-to-end commercial operations of a high-yield mushroom farming and retail business. The system integrates the entire commercial value chain into a single application:

2. Comprehensive 14-Module Functional Specification

Module Code	Module Title	Primary Capabilities & Workflows
MOD-01	**Authentication & RBAC**	Multi-role authentication (NextAuth v5), signed JWT sessions, bcrypt hashing, 40+ atomic permissions across 6 user roles, immutable security audit trail.
MOD-02	**Customer & B2B Profiles**	Customer self-registration, multi-address book with geo-coordinates, B2B restaurant business profiles with GSTIN and credit limits.
MOD-03	**Product Catalog & Taxonomy**	Hierarchical categories, multi-image product showcases, unit selector (KG, 500g, 200g, Pack), B2C vs B2B pricing tiers, shelf-life freshness badges.

Module Code	Module Title	Primary Capabilities & Workflows
MOD-04	**Cart & Dynamic Checkout**	Slide-over cart drawer, free-shipping threshold calculator, address picker, morning/evening delivery slot selector, atomic stock reservation.
MOD-05	**Order Lifecycle Management**	State machine (`PENDING` -> `CONFIRMED` -> `PROCESSING` -> `PACKED` -> `OUT_FOR_DELIVERY` -> `DELIVERED`), invoice and packing slip PDF generation.
MOD-06	**Payment & Reconciliation**	Razorpay / UPI gateway integration, automated webhook signature verification, partial and full refund workflows, COD payment reconciliation.
MOD-07	**Double-Entry Inventory**	Real-time stock ledger, separate finished goods vs raw materials, FIFO tracking, automated low-stock alerts, mandatory reason logging on adjustments.
MOD-08	**Production & Cultivation**	Batch lifecycle tracking (`PLANNED` to `COMPLETED`), spawn-to-substrate ratios, target yield forecasting, Biological Efficiency (BE %) analytics.
MOD-09	**Growing Rooms & Bed Matrix**	Interactive 2D visual room layout (Rooms -> Racks -> Beds), bed health status toggles, contamination logging and quarantine protocols.
MOD-10	**Harvest & Quality Grading**	Mobile-optimized fast harvest entry, multi-flush tracking (Flush #1, #2, #3), quality grade breakdown (Grade A, Grade B, Grade C, Spoilage), auto-stock sync.
MOD-11	**Procurement & Suppliers**	Supplier directory (Spawn, Straw, Bags, Disinfectant), purchase orders, material receipt verification with automatic raw material stock increment.
MOD-12	**Expense Ledger & Accounting**	Categorized expense entries (Electricity, Labor, Rent, Transport, Packaging), receipt photo attachments, payment method tracking.
MOD-13	**Delivery & Fleet Dispatch**	Order dispatch assignment, driver roster, delivery note management, delivery status updates.
MOD-14	**Executive Analytics & P&L**	Real-time Net Profit engine (`Gross Revenue - COGS - OpEx`), yield per bed, cost per kg, customer repeat rates, automated financial export.

3. Database Entity-Relationship (ER) Architecture

The database operates on **MariaDB 11.x** with full referential integrity and optimized indexes:

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4. Key Mathematical Business Formulas

1. **Biological Efficiency (%):**
   $$\text{BE (\%)} = \left(\frac{\text{Total Fresh Harvest Weight (kg)}}{\text{Dry Substrate Weight (kg)}} \right) \times 100$$
2. **Unit Cost of Production (Rs. /kg):**
   $$\text{Unit Cost} = \frac{\text{Spawn Cost} + \text{Straw Cost} + \text{Allocated Power} + \text{Labor}}{\text{Total Harvest (kg)}}$$
3. **Net Profit (Rs.):**
   $$\text{Net Profit} = \text{Gross Revenue} - (\text{COGS} + \text{Operating Expenses} + \text{Logistics} + \text{Taxes})$$

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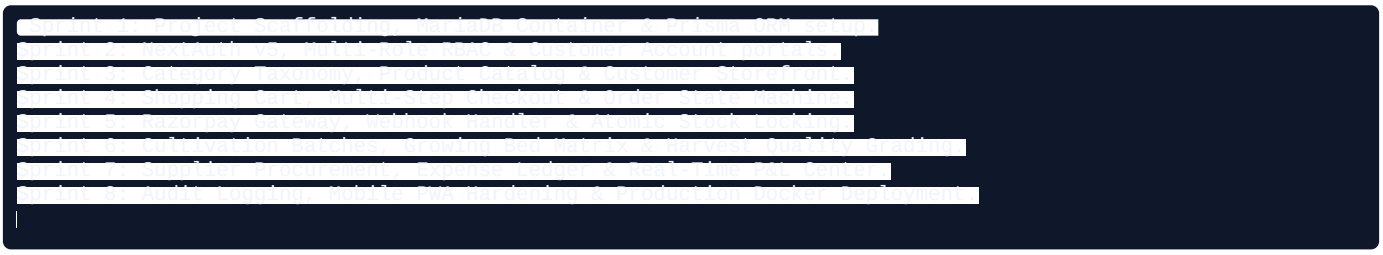
5. Phased Delivery Roadmap Summary